

Integration into LLM

A: The motorcycle is performing a trick in the air.

Early-layer attention may not reflect semantic importance.



LLM Decoder

Transformer Block 2

Transformer Block 1

Tokenizer

Projector

Vision Encoder

Q: What is the motorcycle doing in this image?



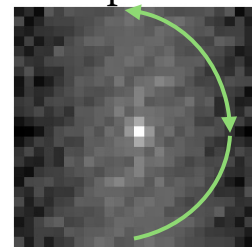
Vision Tokens

Text Tokens

Spectral Evolution

Remove low-frequency components

FFT

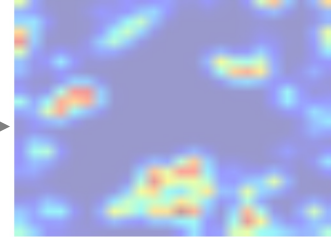


Gaussian High-Pass Filter

iFFT

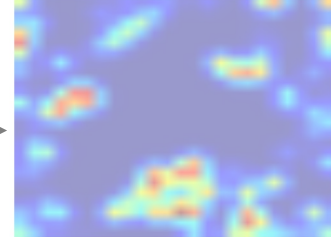
$\mathbf{S}_1$

Subsequent Layer Scores

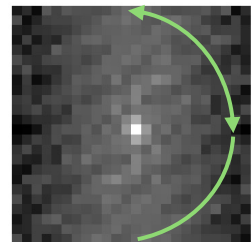


$\mathbf{S}_0$

Reference Layer Scores



FFT

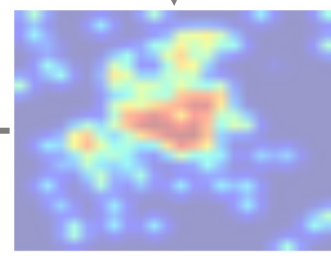


iFFT

Prune Tokens



Tokens with strong spectral evolution are preserved.



$$\left(\frac{|\mathbf{S}_1 - \mathbf{S}_0|}{\mathbf{S}_0 + \text{mean}(\mathbf{S}_0) + \epsilon} \right)$$